Searching the Cognitive Possibility Space:
A Computational Framework for Discovering Learnable Capabilities

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Abstract

Reading is a capability evolution never selected for: culture found an input system that the human brain can learn so well that it becomes automatic. This raises a question that is usually not asked directly: *what else could a fixed learner become, given the right interface and training environment?* We formalise this as **Cognitive Capability Search (CCS)**: an outer loop that searches over interfaces, objectives, input conditions and plasticity schedules for a fixed-architecture learner, scored by how much new world structure becomes cheaply readable (*capability gain*) minus damage to existing abilities (*interference*) — without naming a target capability. In a deliberately small toy (a modular neural network that "develops" with access to only half of its world's latent causes), CCS beats random search under an equal budget (best score 0.50 vs. 0.38; only 0.4% of random environments exceed 0.3), and the environments it selects raise few-shot performance on a held-out battery of tasks that depend on previously inaccessible latents from 0.28 to 0.58–0.60 R^2. The search's most useful finding was one we did not design in: *acquire the new channel while the old sense is silent*. Changing only this one setting moves interference from ≈ 0.01 to 0.35–0.97. Region-level "adaptive plasticity" did not beat global plasticity once the search could exploit this, and the search also exploited a gap in our interference metric. We report these as what a toy can show — a framing and a working method, not a claim about human brains.

1 Motivation: reading as an existence proof

Writing is roughly five thousand years old, far too recent for the brain to have evolved for it. Yet literate adults read automatically, and reading is supported by a reproducible cortical territory in the ventral visual stream. The *neuronal recycling* hypothesis explains this by saying that cultural inventions succeed when they fit pre-existing circuitry closely enough to repurpose it [1]. Sensory substitution and augmentation make the same point from the other side: tactile images can support visual-like perception [2], a vibrating belt can become a usable sense of magnetic north [3], rats can learn to perceive infrared light [4] or navigate by a geomagnetic prosthesis [5], and skin can carry far more information than everyday life asks of it [6].

What these cases share is that *someone guessed the capability first*, then built the interface. Brain–machine interface research already treats the human and the decoder as two co-adapting learners [7, 8], but the task is still fixed in advance. This paper asks a different question:

> *Can we systematically search for capabilities a learner is able to acquire, rather than deciding the target capability in advance?*

In other words, not "how do I teach it X?" but "*what can this learner become?*"

2 Cognitive Capability Search

Ingredients. A learner L with fixed architecture and partly developed weights; a space E of environments; and three measurements taken after L is exposed to $e \in E$:

- **Capability gain** $G(e)$: how much more of the world's latent structure becomes *cheaply* readable from L's internal representation (few-shot linear probes [9]), relative to before.
- **Interference** $I(e)$: loss of performance on abilities L already had, read through their original, frozen readouts.
- **Cost** $T(e)$: how long the learner had to be plastic.

CCS solves

$$e^\star = \arg\max_{e \in E} \; G(e) - \beta\, I(e) - \gamma\, T(e), \tag{1}$$

with no term that names a particular capability. The probe bank is generic — it asks about the world's generative factors, not about a task we hope to see — and a separate held-out task battery, never used by the search, checks whether what was found is useful.

Two loops. The *inner loop* is ordinary learning: L is trained on e's self-supervised objective. The *outer loop* searches over e. Because the outer loop can also choose *which parts of L are plastic and when*, a single environment can implement an *unlock* $\rightarrow$ *learn* $\rightarrow$ *select* $\rightarrow$ *consolidate* cycle, a computational analogue of controlled critical periods [10].

Relation to existing work. CCS borrows its machinery from open-ended search and environment design (POET [11], PAIRED [12], AI-generating algorithms [13]), from novelty and quality-diversity search [14, 15], from intrinsic motivation [16], curriculum learning [17] and meta-learning [18]. The difference is the object of the search. Those methods mostly search for environments that make an *agent* better at a given objective, or generate ever-harder tasks. CCS fixes the learner and asks which *new capabilities* are reachable for it, and at what price in interference with what it already does — the stability–plasticity problem [19, 20] as a constraint on the search space rather than as a nuisance.

3 A toy instantiation

World. Eight latent causes $z \sim \mathcal{N}(0, I_8)$. The learner's innate sense $s \in \mathbb{R}^{16}$ is a fixed nonlinear mixture of $z_0..z_3$ only; $z_4..z_7$ are invisible to it.

Learner. A small modular network (all code is pure NumPy): $h_1 = \tanh(W_1 s + b_1)$ (sensory region R_1), $h_c = \tanh(W_c c)$ (interface region R_c), $h_2 = \tanh(W_{2a} h_1 + W_{2b} h_c + b_2 + \eta)$ (association region R_2, intrinsic noise $\eta \sim \mathcal{N}(0, 0.2^2)$), and a readout R_3 predicting $z_0..z_3$. "Development" trains R_1, R_2, R_3 on the innate task with the new channel silent (innate $\mathrm{R}^2 = 0.968$). The pathway from R_c into R_2 starts weak ($0.15\times$ normal scale), so at birth a new channel reaches the association region only below the noise floor.

Environment space E. Each environment is a genome: *interface* — which latents a new 16-d channel carries (any subset of 8), its code (`linear`, Gaussian-tuned `population`, high-frequency `phase`, or `hashed` lookup) and noise level; *objective* — weights on reconstructing the channel from h_2 and on

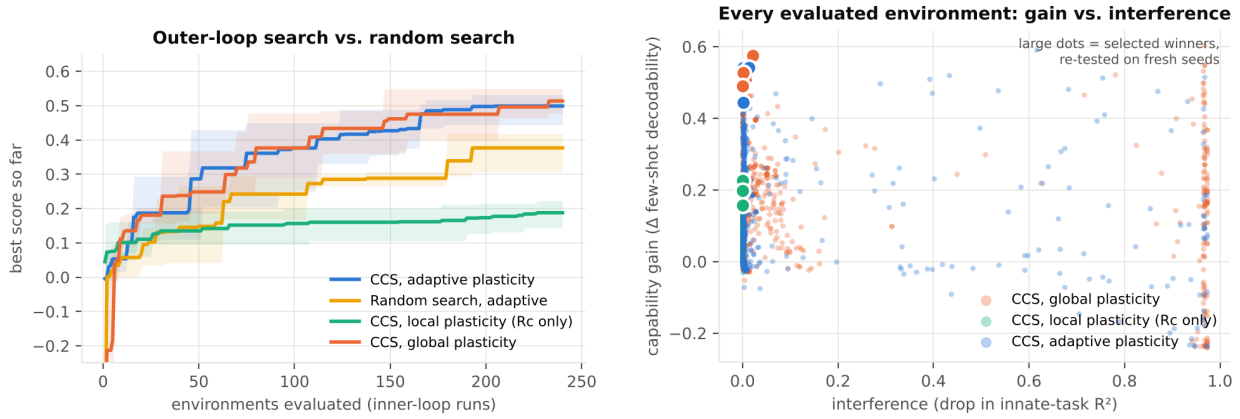


Figure 1: Left: best score found vs. number of inner-loop runs (mean over 3 worlds; band = min–max). **Right:** every environment evaluated by the three CCS searches. Most of the space trades capability for interference; the winners (large dots, re-tested on fresh seeds) sit in the top-left corner.

aligning channel-only h_2 to the representation the innate sense produces (a direct "recycling" signal); *input condition* during learning — channel only, both, or mixed; *plasticity schedule* — two phases, each unlocking any subset of $\{R_1, R_c, R_2\}$ for 0–600 steps; and learning rate — about 1.5×10^8 distinct environments.

Measurements. G is the mean change in few-shot (n=64) ridge-probe R^2 for each of the 8 latents from h_2, under two read-out conditions (channel only; sense + channel). I is the drop in innate-task R^2 through the frozen readout R_3, channel silent. β=1, γ=0.02 per 1200 steps.

Outer loop and baselines. A genetic algorithm (population 20, 12 generations, tournament selection, crossover, mutation, 2 elites) gives 240 inner-loop runs per search. We compare three plasticity regimes — *global* (all regions open for 600 steps), *local* (only R_c), and *adaptive* (schedule searched) — and random search over the adaptive space with the same budget. Two hand-designed environments represent "the obvious first guess": carry all latents in a clean linear code, train by reconstruction with both inputs present, under global or local plasticity. Every result is repeated in three independently generated worlds (different mixing, codes and learner initialisation); winners are re-tested with three fresh inner-loop seeds. The full pipeline runs in about 15 minutes on two CPU cores.

4 Results

Search beats random sampling. With 240 inner-loop runs, GA search reached a best score of 0.50 (adaptive) and 0.51 (global) against 0.38 for random search (Fig. 1, left). Across 720 random environments only 3 (0.4%) exceed 0.3; the GA crossed 0.3 after 30–114 evaluations, random search after 137–192. Re-testing winners on fresh seeds reproduced their scores (Table 1), so this is not a winner's-curse artefact.

The capability that emerges. Fig. 2 shows one selected environment. Before exposure, the learner's association region carries essentially nothing about $z_4..z_7$. After 150 steps in the selected environment, every carried latent is readable from the channel alone ($R^2 \approx 0.92$), and with sense and channel together the hidden latents reach $R^2 \approx 0.65$–0.71 while the visible ones stay at 0.90–0.93. Across all three worlds, the search consistently chose clean `linear` codes carrying 5–7 latents and avoided `phase` and `hashed`

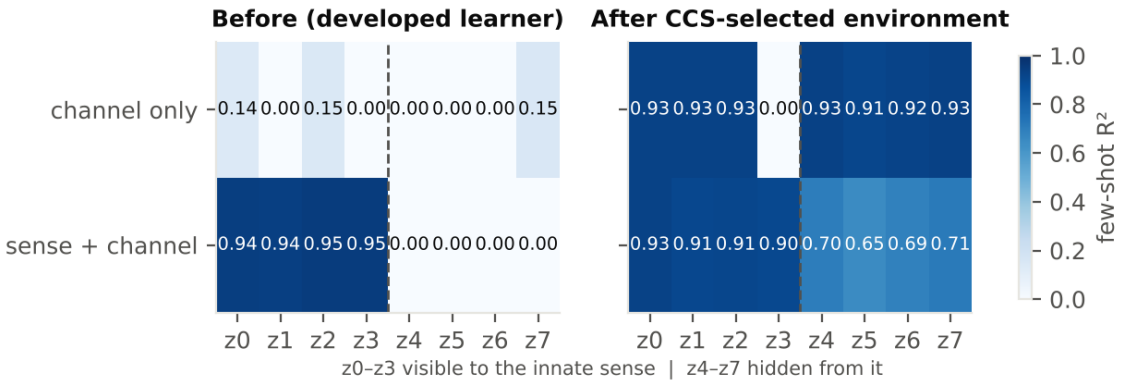


Figure 2: Few-shot decodability of each latent from the association region, before and after the CCS-selected environment (adaptive regime, world 0: carries $z_{0,1,2,4,5,6,7}$, linear code, reconstruction objective, channel-only input, $R_c{+}R_2$ open for 150 steps).

	search objective (fresh seeds)			held-out battery, R^2		
Condition	score	gain G	interf. I	all	visible	hidden
No interface	—	—	—	0.379	0.829	0.283
Hand-designed, global	-0.183	0.661	0.834	0.740	0.785	0.730
Hand-designed, local	0.121	0.131	0.000	0.460	0.819	0.384
Random search, adaptive	0.378	0.390	0.002	0.514	0.796	0.454
CCS, global plasticity	**0.513**	0.531	0.008	0.636	0.792	0.603
CCS, local plasticity	0.184	0.194	0.000	0.403	0.825	0.314
CCS, adaptive plasticity	0.499	0.509	0.005	0.620	0.795	0.583

Table 1: Means over 3 worlds $\times$ 3 inner-loop seeds. The battery is 40 random nonlinear functions of 2–3 latents, read out few-shot from h_2 with sense and channel present; "hidden" tasks depend on at least one of $z_4..z_7$ (33 of 40).

codes, which on average *reduced* decodability (Fig. 4). That interfaces differ sharply in learnability is the toy analogue of the recycling claim that only some cultural codes "fit".

Transfer to tasks the search never saw. On the held-out battery (Table 1, Fig. 3), CCS-selected environments roughly double performance on tasks that need a hidden latent ($0.28 \rightarrow 0.58$–0.60) at a small cost on tasks over visible latents ($0.83 \rightarrow 0.79$). The hand-designed global environment does better still on the battery (0.73) — but it does so by destroying the learner's existing ability: innate-task R^2 falls from 0.97 to 0.11. The battery fits fresh readouts and so cannot see this; the interference term can. The two numbers have to be read together.

What the search found that we did not design: learn while the old sense is silent. All twelve winners chose the *channel-only* input condition. The CCS global winners differ from the hand-designed global environment in small ways (one latent fewer, sometimes more noise or a lower learning rate) and in one large way: they present the new channel *without* the innate sense. Table 2 isolates this: holding everything else fixed and changing only the input condition keeps the gain (0.51–0.57) but moves interference from ≤ 0.01 to 0.35–0.97. Mechanistically, with $s = 0$ the sensory weights W_1 receive exactly zero gradient and h_1 is constant, so the pressure to change is routed through the interface pathway (in world 0, the relative change in W_{2b} is $\approx 6\times$ its initial norm, in W_{2a} only 0.19); we have not fully characterised why the remaining

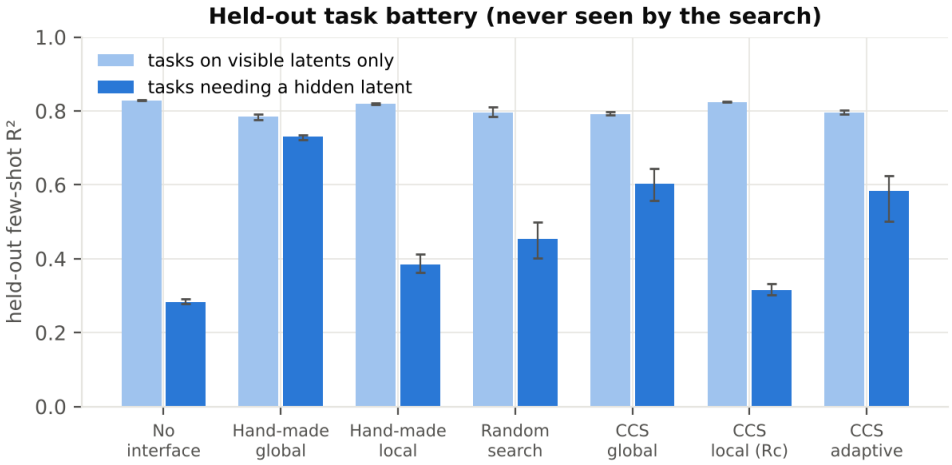


Figure 3: Held-out battery, mean over 3 worlds (error bars = min–max).

Winner	input during learning	gain	interference	score
CCS, global	channel only (as found)	0.531	0.008	**0.513**
	mixed	0.571	0.968	−0.407
	both	0.553	0.827	−0.284
CCS, adaptive	channel only (as found)	0.509	0.005	**0.499**
	mixed	0.546	0.763	−0.221
	both	0.510	0.350	0.157

Table 2: Ablation: only the input condition is changed. Means over 3 worlds × 3 seeds.

changes to R_2 spare the innate readout. The pattern is loosely reminiscent of the recruitment of deprived cortex by Braille in blind readers [21], but we make no claim beyond the toy.

Adaptive plasticity: a negative result, with a caveat. We expected region-level control ("open R_2 briefly, then freeze") to beat global plasticity. It did not: 0.499 vs. 0.513, within the spread across worlds. Once input gating was available, it did the work that plasticity gating would otherwise do. The caveat is in Table 2: when the protective input condition is removed, the adaptive winners' shorter, partial schedules keep interference at 0.35 against 0.83 for global plasticity. Plasticity control looks like a second line of defence rather than the main one — suggestive only, with three worlds. Purely *local* plasticity (only the interface region) was the clearest failure: R_c alone cannot amplify a pathway into R_2 that starts below the noise floor, so gains plateaued at ≈ 0.19. A new capability needed the association region to change.

The search exploited our metric. Interference was measured with the new channel silent. With the channel switched *on*, CCS winners' innate-task R^2 drops from 0.97 to 0.88 (vs. 0.31 for the hand-designed global environment and 0.96 for local plasticity). Nothing in the score penalised this crosstalk, so the search did not avoid it. This is the general lesson of any objective-driven search, and for CCS it is a design requirement: the interference term must cover every context in which old abilities will be used.

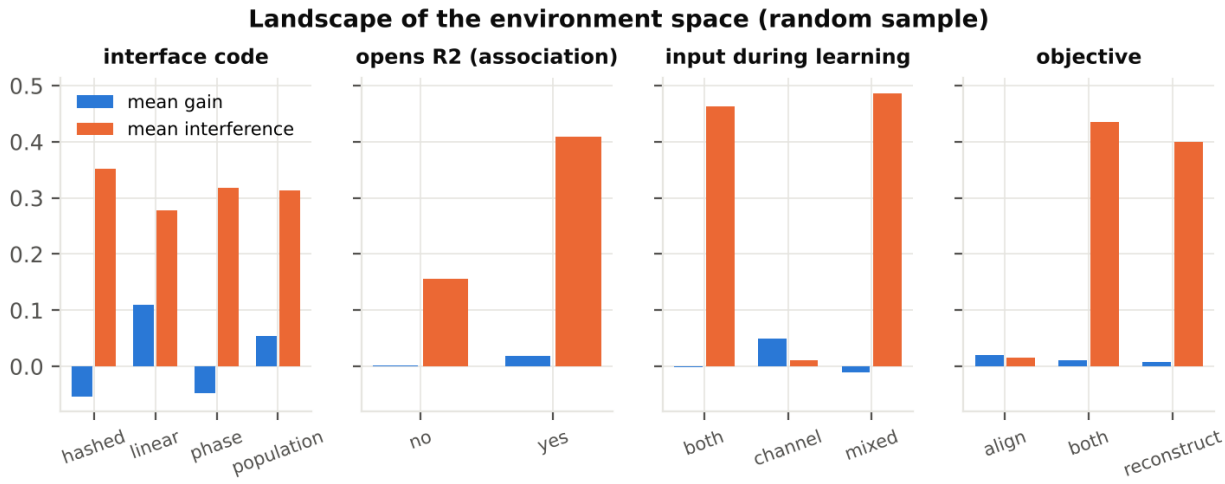


Figure 4: Mean gain and interference per gene value, over 720 random environments (an unbiased sample of E).

5 What this does and does not show

It shows that, for a fixed learner, the space of interfaces and training regimes can be searched *for capabilities* rather than *for a task*; that such search finds far better regions of that space than random sampling; that what it finds transfers to tasks it never saw; and that it can surface non-obvious training principles (here, input gating) that a designer can then test directly. It also shows the method's main hazard: it optimises exactly the interference you measure.

It does not show anything about human brains. The learner is a three-region $\tanh$ network; the "capability" is linear decodability of a synthetic world's factors; three worlds and 240 evaluations are small. The probe bank, though generic, is still a choice — capabilities that are not functions of z (new kinds of computation rather than new information) are invisible to it.

6 Why this might matter

The near-term version of CCS is practical: searching the design space of sensory-substitution devices, BCIs and training curricula with an explicit interference budget, using models of the learner fitted to behavioural or neural data as the inner loop. The far version is a question: are there cognitive modalities with no cultural analogue yet — for example, a brain that has learned to treat another person's neural state as an ordinary perceptual input — sitting inside the constraints of the human brain, undiscovered because nobody has built the right interface or training environment? We do not claim this toy says anything about that. We claim it is the right shape of question, and that it can be posed as a search.

Reproducibility. Code, results and figures: `python run_experiments.py && python run_ablation.py && python make_figures.py` (NumPy and Matplotlib only). Developed with AI assistance (Claude) for code and drafting; all numbers come from the released code.

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